

```
!pip install pymilvus
```

```
Requirement already satisfied: pymilvus in /usr/local/lib/python3.12/dist-packages (2.6.2)
Requirement already satisfied: setuptools>69 in /usr/local/lib/python3.12/dist-packages (from pymilvus) (75.2.0)
Requirement already satisfied: grpcio!=1.68.0,!=1.68.1,!=1.69.0,!=1.70.0,!=1.70.1,!=1.71.0,!=1.72.1,!=1.73.0,>=1.66.2 in /usr/
Requirement already satisfied: protobuf>=5.27.2 in /usr/local/lib/python3.12/dist-packages (from pymilvus) (5.29.5)
Requirement already satisfied: python-dotenv<2.0.0,>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from pymilvus) (1.1.1)
Requirement already satisfied: ujson>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from pymilvus) (5.11.0)
Requirement already satisfied: pandas>=1.2.4 in /usr/local/lib/python3.12/dist-packages (from pymilvus) (2.2.2)
Requirement already satisfied: typing-extensions~4.12 in /usr/local/lib/python3.12/dist-packages (from grpcio!=1.68.0,!=1.68.1
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.2.4->pymilvus) (2.0.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.2.4->pymilvus)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.2.4->pymilvus) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.2.4->pymilvus) (2025.2)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas>=1.2.4-
```

```
from pymilvus import connections, FieldSchema, CollectionSchema, Collection, DataType
```

```
# Public endpoint from your cluster
```

```
endpoint = "https://in03-2c505f19b878ebf.serverless.aws-eu-central-1.cloud.zilliz.com"
```

```
# Your token in the format "user:password" (e.g. db_admin:YOUR_TOKEN)
```

```
token = "8ef475bb3208de7308647cdd5a99f0d86cfdb8c14d7197e0ea532ca417eed5790573c33a0b10b8c0de7039415a9375766d0f9e15"
```

```
# 1. Connect using URI + token
```

```
connections.connect(
    alias="default",
    uri=endpoint,
    token=token
)
```

```
print("✅ Connected to Zilliz Cloud (Serverless)")
```

```
✅ Connected to Zilliz Cloud (Serverless)
```

```
# 2. Create a collection
```

```
fields = [
    FieldSchema(name="id", dtype=DataType.INT64, is_primary=True, auto_id=True),
    FieldSchema(name="embedding", dtype=DataType.FLOAT_VECTOR, dim=128),
]
```

```
schema = CollectionSchema(fields, description="DemoCollection")
```

```
collection = Collection("demo_collection", schema=schema)
```

```
print("✅ Collection created")
```

```
✅ Collection created
```

```
# 3. Insert sample vectors
```

```
import numpy as np
```

```
vectors = np.random.random((5, 128)).tolist()
```

```
collection.insert([vectors])
```

```
print("✅ Inserted 5 vectors")
```

```
✅ Inserted 5 vectors
```

```
# 3b. Create an index on the vector field
```

```
index_params = {
    "metric_type": "L2",
    "index_type": "IVF_FLAT",
    "params": {"nlist": 128}
}
```

```
collection.create_index(
    field_name="embedding",
    index_params=index_params
)
```

```
print("✅ Index created")
```

```
✅ Index created
```

```
# 4. Search
```

```
collection.load()
```

```
query = [vectors[0]]
```

```
results = collection.search(
    data=query,
    anns_field="embedding",
    param={"metric_type": "L2", "params": {"nprobe": 16}},
    limit=3
)
```

```
)  
  
print("🔍 Search results:")  
for hit in results[0]:  
    print(f"ID = {hit.id}, Distance = {hit.distance:.4f}")
```

```
🔍 Search results:  
ID = 461027866603334443, Distance = 0.0000  
ID = 461027866603334446, Distance = 20.3280  
ID = 461027866603334444, Distance = 20.9005
```

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